1. Rho cycle windows in bitop/v1, and the rho Ir/gate spread

research/rho_cycle_window_bitops_20261107 · 2026-11-07 · branch claude/rho-cycle-window-bitops · base a2be660914 · class: accounting. All values are unverified assertions [pcc:unverified].

1.1. The cycle windows are charged

The default hyperelliptic rho reference (DistinguishedPointsNegationFullWidthCollideFirst) keeps a 32-entry recent-hash window whose work was not in the bitop/v1 tally; ecbench's tuned rho walker (ic_boundary TunedWalker, its lanes, Brent under WalkPolicy::v2) had the same gap with its 16-entry ring. Now charged: one word_op per hash compare of the scan and per push test; the escape step's RNG draw, branch index and two modr_add (as the gap-limit escape); the collide-first solve and check moved to linear_algebra and verification. ecbench carries its counts as the search phase's work_native (cycle_window_word_ops, brent_word_ops), so committed records replay and price unchanged. Memory, clearing (control) and the passive instrument window are not charged (docs/units/BITOPS.md, 2026-11-07).

1.2. Which published ratios used the default walk

None: the hec_bitop_20261020 rows and the leaderboard section sourced from them used the full-width walk (no window); rounds 14–15 compare rho with rho in walk group operations; round 16 is ecbench GAE. The leaderboard and dashboard are unchanged. To size the change, the rho arm of all 18 hec_bitop rows was re-derived with examples/hec_rho_window_reprice.rs (same instances, 40 seeds from 0xC0FFEE, counts only). Control: the full-width re-derivation reproduces all 18 committed rho means exactly. Every run verified.

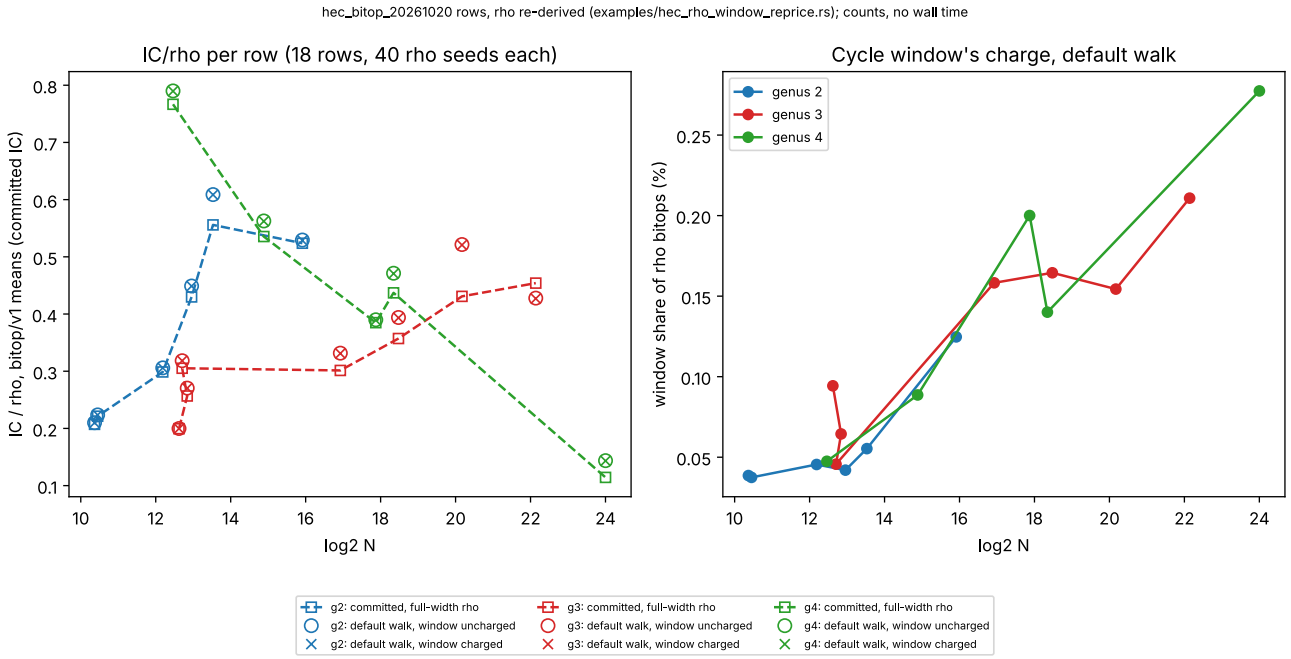


Figure 1: Left: IC/rho per row, committed (full-width rho) and against the default walk with the window uncharged and charged. Right: the window's share of the default walk's bitop/v1 total. 18 rows, 40 rho seeds per row, committed IC means; raw/collide-first.jsonl.

g	p	N	committed IC/rho (full-width)	default walk, window uncharged	default walk, window charged	window share %
2	41	1321	0.2071	0.20999	0.20991	0.039
2	61	1399	0.2213	0.22362	0.22354	0.037
2	101	4663	0.2987	0.30587	0.30574	0.045
2	151	7949	0.4299	0.44897	0.44878	0.042
2	211	11813	0.5558	0.60883	0.60849	0.055
2	251	61667	0.5238	0.52938	0.52872	0.125
3	23	6299	0.1995	0.19985	0.19966	0.094
3	31	7333	0.2567	0.27048	0.2703	0.064
3	41	6679	0.3052	0.31836	0.31822	0.046
3	61	124459	0.3014	0.33158	0.33106	0.158

3	101	364747	0.3572	0.39393	0.39328	0.165
3	151	1180351	0.4309	0.5215	0.5207	0.154
3	211	4620611	0.454	0.42782	0.42692	0.211
4	17	5639	0.7667	0.79005	0.78968	0.047
4	23	30223	0.5355	0.56271	0.56221	0.089
4	31	239753	0.3849	0.38994	0.38916	0.2
4	41	333041	0.437	0.47129	0.47063	0.14
4	61	16790591	0.1144	0.14367	0.14327	0.277

The window costs 18–31 word operations per walk group operation against 10^5–10^6 gates per Jacobian step: 0.04–0.28% of the walk; every IC/rho ratio falls by the same 0.04–0.28%. Aside, not acted on: the default walk is up to 20% cheaper than the full-width walk the published rows used, so the genus-4, p = 61 row reads 0.143 against it instead of the published 0.114.

1.3. The rho Ir/gate spread of ic-acct-7

Hypothesis: fixed per-solve work that bitop/v1 does not charge dominates short rho walks. It holds. Committed audit-ir (session binary bf8196c2, seq 9 and 11): Ir = 2.335e6 + 0.0869·gates. Local ladder, ecbench audit-ir on all 12 rho records of the spec with binary 4ff2214e: Ir = 1.441e6 (±1.9e3) + 0.08704 (±0.0007)·gates, R^2 = 0.9993. callgrind_annotate names the fixed work, identical in every record: SignedFrobeniusClasses::new ≈ 605,300 Ir (FrobeniusPowers::new 493,698; FrobeniusCanon::new ≈ 110,800), the per-solve build of the Frobenius tables that canon and as_solve are priced as if precomputed; and full_price::reprice ≈ 778,000–786,000 Ir, the harness re-pricing its report (JSON parse and SHA-256 of full_prices.json) inside the bracketed solve. Without them Ir = 5.7e4 + 0.0859·gates (R^2 = 0.9997) and Ir per gate is 0.095–0.133 on every record, against 0.31–1.22 before.

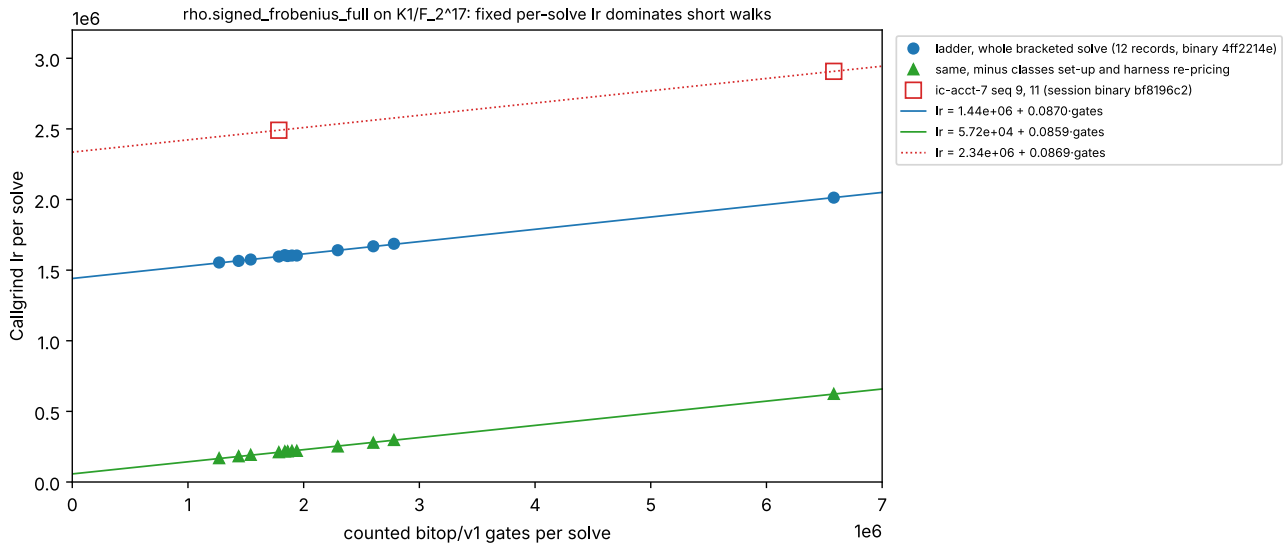


Figure 2: Callgrind Ir against counted bitop/v1 gates, rho.signed_frobenius_full on K1/F_2^17 (spec koblitz-symmetrised-n17). Blue: 12 ladder records, whole bracketed solve; green: the same minus the two named fixed parts; red: the two committed ic-acct-7 records. ir_spread/fit-*.json.

Recorded as known constants, not charged: the re-pricing is instrumentation (move it out of the bracket or subtract it in audit-ir); the classes build is per-curve set-up that bitop/v1 omits for every Frobenius-folded rho, to be charged with its own counter as the next step. Open: the rho marginal, 0.086 Ir per gate, is below the IC arm's 0.30–0.32; either IC phases still miss work or the binary-field canon/as_solve weights over-price the rho walk. Two IC records do not separate the two.

1.4. Limits

Counts and Callgrind only, shared L0 host, no wall time. No AWS identity here, so nothing ran on isolab. Descriptive; no row promoted, no VALIDATION.json, no leaderboard or benchmarks row. Resume checkpoint not pushed (no upload token on the host).